

Malignant Melanoma: Epidemiology, Pathogenesis, and Clinical Staging

Global Oncology Consortium

BACKGROUND: Malignant melanoma is the most aggressive form of skin cancer, arising from the malignant transformation of melanocytes. Its incidence has been rising globally over the past few decades.

PATHOGENESIS: The development of melanoma involves a complex interplay of genetic mutations (such as BRAF V600E, NRAS, and KIT) and environmental factors, predominantly ultraviolet (UV) radiation. Brain metastasis occurs in up to 50% of advanced cases, frequently presenting with severe headaches.

DIAGNOSIS & STAGING: The ABCDE criteria remain the clinical standard for early detection. The AJCC staging system relies on tumor thickness (Breslow depth), ulceration, and lymph node involvement to determine prognosis.